Tobin Carlton Van Buizen

Mathematics Honours and Statistics, Australian National University.
Research interests: equivariant stable homotopy theory & THH/TC.
Accelerated entry: commenced university at 15; completed Honours year at 20.

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Education

The Australian National University — Canberra, Australia
Bachelor of Mathematical Sciences (Honours) — 2021 – 2025

- First Class Honours
- Supervisor: Dr. Vigleik Angeltveit.
- Research area: Equivariant Stable Homotopy Theory
- Descriptive course list (upper-division/graduate): Available here.

The Australian National University — Canberra, Australia
Bachelor of Statistics — 2021 – 2024

- Descriptive course list (upper-division/graduate): Available here.

Australian Mathematical Sciences Institute — Canberra, Australia
Summer School 2024 — 2023 – 2024

- Descriptive course list: Available here.

Research Projects

Honours Thesis: Twisted Topological Hochschild Homology
Australian National University — 2025.01 – 2025.11

- Supervisor: Dr. Vigleik Angeltveit.
- Developed a twisted model of p-cyclotomic spectra and twisted TC for C_n-ring spectra.
- Provided a Nikolaus-Scholze style theorem for computing twisted TC.
- Provided an identification between norms and twisted cyclic bar constructions.

Odd Primary Hopf Invariant One
Australian National University — 2025.02 – 2025.06

- Surveyed odd-primary refinements and constraints on Hopf invariant one.

Lévy Integration
Australian National University — 2024.08 – 2024.11

- Constructed Lévy-Itô integration for general Lévy processes and compare with Itô integration.
- Provided worked examples for compound Poisson and variance-gamma processes.

Applications of Differential Forms to Homotopy Theory
Australian National University — 2024.08 – 2024.11

- Surveyed rational homotopy theory and differential graded algebras.

Kervaire Invariant One: C_2-Equivariance and KR-Theory
Australian National University — 2024.02 – 2024.06

- Expository project connecting C_2-equivariant techniques to the Hill-Hopkins-Ravenel approach.

UMD Spaces
Australian National University — 2023.08 – 2023.11

- Expository project on spaces with unconditional martingale differences, and applications to stochastic calculus.

Teaching

Demonstrator | The Australian National University 2023.07 – Present

- **MATH1005:** Discrete Mathematical Models (2023.07 – 2024.11) - ran workshops, marked assignments, marked exam and wrote solutions.
- **MATH1115:** Advanced Mathematics and Applications 1 (2026.02 – 2026.06) - running workshops, marking assignments and marking exams.
- **MATH2222:** Introduction to Mathematical Thinking (2025.02 – 2025.06) - ran workshops, marked assignments and marked exam.
- **MATH4204:** Algebraic Topology (Honours) (2025.07 – Present) - marked assignments.
- **MSI Drop-In Help Centre** MSI Drop-In Help Centre (2026.08 – Present) - answering questions from students in all MATH courses at ANU, primarily first and second year.

Talks

THH with a "Twist" - Honours Conference, ANU 2025.11

What is... G-stuff? - What is...? Seminar, ANU 2025.08

Odd Primary Hopf Invariant One - VBKT Seminar, ANU 2025.06

The Scariest Maths Topics (in 8 minutes) - Maths in the Pub 2025.05

Lévy Integration - Stochastic Analysis & Financial Applications 2024.11

C_2-Equivariance and KR-Theory - SHT Seminar, ANU 2024.06

Service

Co-organiser, "What is...?" student seminar, ANU (2025)

Work Experience

Evolve Capabilities Canberra, Australia
Data Analyst 2026.01 – Present

Australian Public Service Canberra, Australia
Configuration Management & Reporting Service Delivery Manager 2023.02 – 2026.01

Skills

Programming: `Python`, `R`, `MATLAB`, `HTML`, `CSS`; `SQL`, `Excel`, `PowerBI`, `PowerQuery`

Typesetting: LaTeX (BibTeX, TikZ, TikZ-cd)

Languages: English (native); some Chinese

Awards

- Caltex Best All Rounder Award - Presented in recognition of excellence in academic, leadership, sporting and community service activities.
- Member of MENSA Australia.